Lefkios Mavroudi

mavroudi.lefkios03@gmail.com | LinkedIn | GitHub

Education

Imperial College London — *London, UK*

MSc Computing (Artificial Intelligence and Machine Learning) — *Sep 2026 – Sep 2027 (expected)*

- Recipient of the Safe Bulkers Scholarship, one of 2 postgraduate recipients for 2026–27.

University of Cyprus — *Nicosia, Cyprus*

BEng Electrical and Computer Engineering — *2022 – 2026*

- GPA: 8.46/10; ranked 2nd/39 in Computer Engineering and 4th/83 in the ECE department.
- Thesis (10/10): *Adversarial Robustness of Edge-Deployed Object Detectors in Mobile Robotics* — awarded Best Senior Design Project (department's best Computer Engineering thesis). GitHub.
- Second-highest academic performance distinction in Computer Engineering: 2023–24, 2024–25, 2025–26.

Experience

CYENS Centre of Excellence — *Jun 2026 – Jul 2026*

AI Research Engineer Intern — *Nicosia, Cyprus*

- Implemented and evaluated a MedSAM2 adaptation for right-ventricle segmentation in echocardiography, and built an automated PyTorch pipeline converting transfer-model predictions into bounding-box prompts.
- Improved locked-cohort mean Dice from 0.8463 to 0.8918 (+4.55 points) across 41 held-out cases through prompt ablations and staged decoder fine-tuning, improving 74/82 annotated frames.

KIOS Research and Innovation Centre of Excellence — *Jun 2025 – Jun 2026*

Researcher — Adversarial ML and Edge AI — *Nicosia, Cyprus*

- Developed and validated a PyTorch Expectation-over-Transformation adversarial-patch pipeline, reducing YOLOv8n mAP@0.50 from 80.6% to 49.2% and increasing high-confidence false positives by $\sim$40$\times$; demonstrated cross-scenario and black-box transfer across YOLOv8, YOLOv5, NanoDet+, and RT-DETR-L.
- Evaluated preprocessing defences, with Local Gradient Smoothing reducing false-positive flooding by 63.6%; deployed and profiled the system in CARLA-GeAR and on NVIDIA Jetson AGX Orin, measuring latency, FPS, power, and energy overhead.
- **First author** of *RAPTOR: Robustness Assessment of Patch Transfer and Operational Risk in Edge Object Detectors for Mobile Robots*, presented at IEEE COINS 2026.

Deloitte — *Jun 2024 – Jul 2024*

Cyber Risk Intern — *Nicosia, Cyprus*

- Performed ethical-hacking assessments and EU compliance exercises, documenting findings and prioritising remediation.

3AE Health — *Jun 2023 – Jul 2023*

AI Software Engineering Intern — *Nicosia, Cyprus*

- Co-developed a traffic-analytics system in a two-person team, integrating YOLOv8s vehicle detection, multi-object tracking, and five-minute traffic-flow aggregation for forecasting and pattern analysis.
- Built a stacked-GRU traffic-flow forecasting model, achieving a 0.041 normalised validation MAE.

Selected Projects

LLM Post-Training: Process vs Outcome Rewards — *Aug 2026*

PyTorch | Transformers | TRL/GRPO | PEFT/LoRA | OpenAI API | GitHub

- Built LLM post-training and evaluation pipelines comparing outcome rewards with human- and GPT-4o mini-scored process rewards; fine-tuned Qwen2.5-7B using GRPO, TRL, and LoRA on Apple MPS.
- Human process supervision achieved 94.5% versus a 91% baseline on a 200-problem GSM8K best-of-8 evaluation; in-domain training achieved 54.5% versus 49% on a 200-problem MATH-500 subset (Level 5: 8% $\rightarrow$ 20%).

MIPS32 Processor Simulator — *Feb 2025 – Apr 2025*

C++ | Computer Architecture | Pipelining | GitHub

- Implemented cycle-accurate single-cycle, multi-cycle and pipelined MIPS32 simulators in C++, including datapaths, hazard detection, forwarding and control-flow handling; awarded 10/10.

Adversarial Person-Hiding Patches — *Aug 2025 – Sep 2025*

Python | PyTorch | YOLOv8 | GitHub

- Optimised printable person-hiding patches for a live Researchers' Night 2025 demonstration using top-K suppression, TV/L2 regularisation, JPEG compression, blur, and LGS.

Technical Skills

Languages: C++, Python, C, Java, SQL, MATLAB

Systems and Tools: Linux, Docker, Git, CUDA, Jetson AGX Orin, ROS 2, Nav2, LiDAR, Isaac Sim

Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, TRL/GRPO, PEFT/LoRA, MLX, Ultralytics YOLO, OpenCV, scikit-learn